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ECONOMICHE E SOCIALI

Statistical Methods for Machine Learning class

ASSIGNMENT 5 CLUSTERING AND INITIALIZATION

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Chapter 1

Introduction

In this project I implement the unsupervised algorithms K-means and K-means++, specifically by examining the internal structure of both algorithms and comparing their differences and resulting outcomes. They are tested on Fashion-MNIST and on synthetic Gaussian blobs reduced to the same final dimension. The goal is to assess clustering quality after mapping clusters to the true labels, and to observe the effect of the initialization strategy, from random selection to selecting each new centroid with probability proportional to the squared distance to the nearest existing centroid.

So through this project I am researching the widely known limitation of K-means of its sensitivity to initialization. Since the K-means objective is non-convex as the algorithm can converge to a poor local minimum depending on the initial placement of the centroids, so each experiment is run several times to observe the variability of the results. K-means++ is expected to reduce this sensitivity by producing a lower inertia, meaning points that are more compactly grouped around their centroids (Arthur and Vassilvitskii, 2007). A central question of this project is whether that lower inertia actually translates into better clustering with respect to the true underlying label distribution (Equation 1). Could it be a good correlation with the performance of unsupervised learning generally?

Chapter 2

Dataset

The first real-world dataset used in this project is Fashion-MNIST, chosen as a representation of a multiclass, high-dimensional data distribution. After standard scaling and normalization, the original $28 \times 28 = 784$ dimensional images are reduced to 50 dimensions via random projection, which approximately preserves pairwise Euclidean distances (Johnson–Lindenstrauss lemma) and keeps the K-means distance computations meaningful (Dasgupta and Gupta, 2003).

The challenge with Fashion-MNIST is that its true label classes, being high-dimensional images, have a dense, overlapping data spread. On one hand, this complexity helps to research inertia and its varying spread in each random iteration; on the other hand, it helps research the actual power of the chosen initialization approaches. This complexity can also increase the likelihood of so-called empty clusters. So with the random initialization approach, K-means has a higher chance of falling into a poor local minimum, whereas K-means++ offers a good way to address this by forcing each new centroid to be chosen with probability proportional to D^2 .

On the other hand, the synthetic Gaussian blobs, sampled from a Gaussian mixture are an ideal scenario of spherical clusters that match the assumptions of K-means and have known ground-truth centers. This makes the checking would minimizing inertia until convergence correlate with better recovery of the true underlying distribution.

Chapter 3

Methodology

K-means is an unsupervised clustering algorithm that partitions n data points into k clusters without using any labels. It is an iterative procedure, known as Lloyd’s algorithm, that alternates between two steps: an assignment step, where each point is assigned to the nearest centroid by Euclidean distance, and an update step, where each centroid is recomputed as the mean of the points assigned to it ¹. These steps repeat until convergence, progressively minimizing the inertia objective (Equation 1) . The inertia is non-convex, meaning it has multiple local minima rather than a single global minimum. Because the assignment of data points relies strictly on initial centroid placement, standard K-means—which selects these initial centroids uniformly at random—often traps the algorithm in a poor local minimum. K-means++ resolves this by modifying only the initialization step; the assignment and update loop remains identical. Instead of purely random selection, K-means++ spreads the initial centroids out: the first centroid is chosen at random, and each subsequent one is selected with a probability proportional to its squared distance to the nearest already-chosen centroid. This mathematically forces the starting points to be widely distributed, as detailed in Algorithm 2.

Algorithm 1 K-means Clustering

Require: Data X , number of clusters $k = 10$

- 1: **Initialize:** Randomly select k data points as the initial centroids.
▷ `initialize_centroids(X, k)`
 - 2: **repeat**
 - 3: Measure the distance between each data point and the centroids.
▷ `Euclidean_distance(X, centroids)`
 - 4: Assign each data point to the nearest centroid.
▷ `assign_clusters(distances)`
 - 5: Calculate the mean of each cluster.
 - 6: Update the centroids to these means (new centroids for each cluster).
▷ `mean_image_of_cluster(X, labels, k) / center_of_cluster(X, labels, k)`
 - 7: **until** convergence
-

Algorithm 2 K-means++ Initialization

Require: Data X , number of clusters k

- 1: Randomly select the first centroid from the data points.
▷ Call `random_init_centroids` with $k = 1$ for this first step only.
 - 2: **for** $i = 2$ **to** k **do**
 - 3: For each data point x_n , compute its squared distance to the nearest existing centroid:
$$D(x_n) = \min(\|x_n - c_1\|^2, \|x_n - c_2\|^2, \dots, \|x_n - c_{i-1}\|^2)$$
 - 4: Select the next centroid c_i from X with probability proportional to $D(x_n)$:
$$P(x_n) = \frac{D(x_n)}{\sum_n D(x_n)}$$

This ensures that points farther from existing centroids have a higher chance of being selected.
 - 5: **end for**
 - 6: **return** the k selected centroids.
-

$$\text{Inertia} = \sum_{i=1}^n \|x_i - \mu_{c(i)}\|^2 \quad (1)$$

Chapter 4

Experimental Results

General insight is laid down on the mathematical goal of K-means, creating spherical clusters of nearby points, which does not perfectly align with the true human labels for both the real-world and the synthetic data. tracked inertia to measure cluster compactness, initially expecting lower inertia to yield higher accuracy. Owing to the synthetic data, it is safe to conclude that the accuracy mostly depends on how visually clean and well-spread the clusters are (cluster centers drawn uniformly in the range $\text{low} = -5$ to $\text{high} = 5$ in each dimension, with a standard deviation of 3.0 controlling how far the points spread around each center), which directly correlates with better accuracy (Figure 1). So inertia and true-label accuracy do not move together.

The good thing of testing on synthetic blobs that it effectively shows the high variability of simple K-means. Even on mathematically separated clusters, purely random initialization frequently traps the algorithm in suboptimal configurations, resulting in a wide spread of final inertia values across different seeds. In contrast, K-means++ resolves this by producing a tightly concentrated inertia range, consistently finding a better, lower-inertia local minimum. 2.

Simple K-means produces a wide spread of final inertia across seeds, while K-means++ is concentrated in a much tighter range. However, in Fashion-MNIST both methods exhibit a similar overall range of inertia across the tested seeds and share a distinct low point at seed 789. This indicates that in complex, non-spherical feature spaces, spreading out the initial centroids with K-means++ does not guarantee a

massive reduction in absolute inertia compared to standard random initialization.

Fashion-MNIST forms a dense, overlapping cloud, reduced from 784 to 50 dimensions, with no clearly separated groups, so the true classes are not spherical and well-separated in the way K-means requires (Figure 1). This overlap of the classes based on algorithm’s geometric assumptions, rather than the projection itself, leads directly to incorrect labeling. K-means++ gives a modest improvement, most visibly on some classes.

The measurable advantage of K-means++ is therefore reduced variability across initializations, most visible on the synthetic data, rather than a large reduction in the absolute inertia.

Chapter 5

Conclusion

The conclusion for the optimization of K-means++ is that its centroid seeding positively impacted the inertia, however it did not make much difference to the accuracy. As with any unsupervised learning algorithm, it has no intention of matching the true labels. Across the five seeds, K-means++ produced a much tighter and generally lower inertia, most clearly on the synthetic data.

Another goal of this project was to research the widely known limitation of K-means, its sensitivity to initialization. However, honestly, in my case, even though K-means++ showed a lower inertia, meaning points that are closer to their centroids, this did not directly reflect in better clustering with respect to the true underlying label distribution.

Bibliography

- Arthur, D. and Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding. In *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA '07)*, pages 1027–1035, Philadelphia, PA, USA. Society for Industrial and Applied Mathematics.
- Dasgupta, S. and Gupta, A. (2003). An elementary proof of a theorem of Johnson and Lindenstrauss. *Random Structures & Algorithms*, 22(1):60–65.

Appendix: Figures

PCA Visualization - All Datasets & Methods

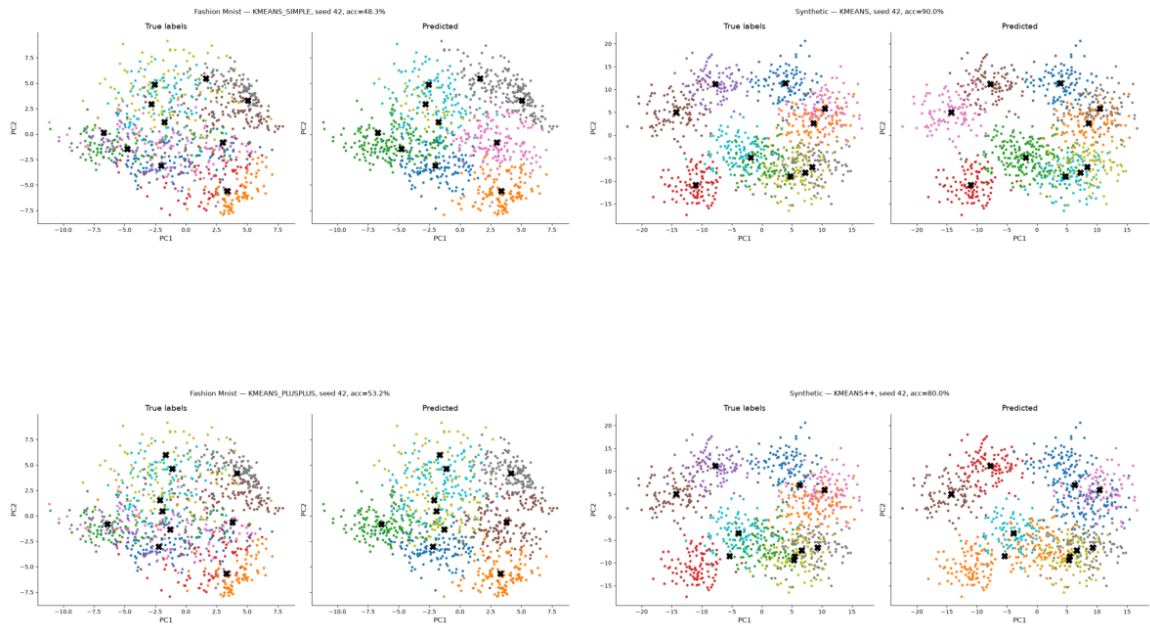


Figure 1: PCA visualization of true labels versus predicted clusters for both datasets and both methods.

Inertia and Variability - All Datasets & Methods

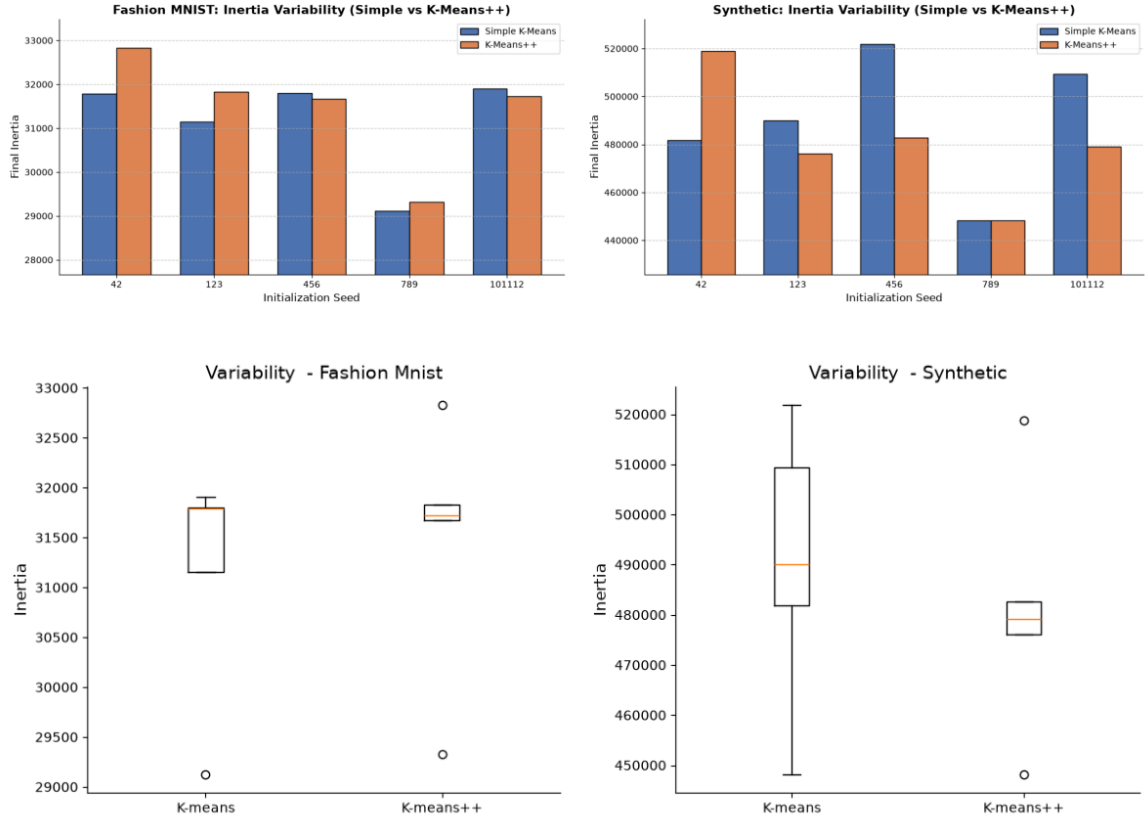


Figure 2: Final inertia across five initialization seeds for simple K-means and K-means++.

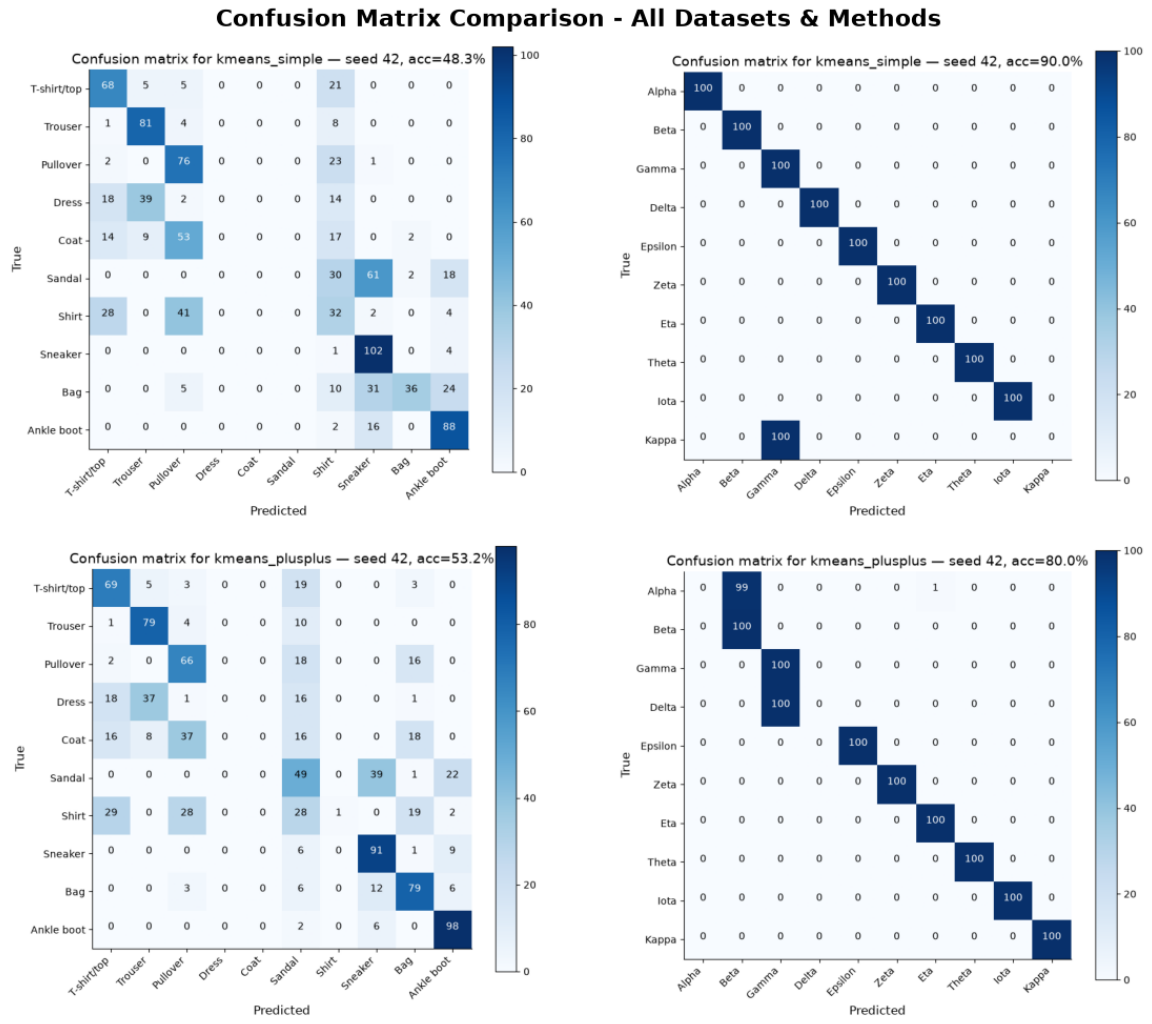


Figure 3: Confusion matrices at seed 42 for both datasets and methods.